

Abaid-ur-Rehman

Machine Learning Engineer | AI Engineer | Python Developer

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SUMMARY

Machine Learning Engineer and Python Developer with hands-on experience building **AI, Machine Learning, Computer Vision, Generative AI, LLM, and RAG applications**. Skilled in Python, Flask, TensorFlow, Scikit-learn, OpenCV, FAISS, and modern LLM APIs, with experience developing intelligent, full-stack solutions for real-world problems. Passionate about designing practical AI systems and continuously exploring emerging AI technologies.

EDUCATION

Bachelor of Science in Artificial Intelligence <i>NFC Institute of Engineering and Fertilizer Research, Faisalabad</i>	2023 – Cont.
Intermediate (Pre-Engineering), KIPS College, Faisalabad	2021 – 2023
Matriculation (Science), Allied School, Faisalabad	2019 – 2021

PROJECTS

StudyMind- AI, RAG / Generative AI / LLM

Technologies: Python, Flask, SQLAlchemy, FAISS, PyMuPDF, Google Gemini, OpenAI, Anthropic APIs, Flask-Login, Flask-Mail, JavaScript, HTML5, CSS3

- Developed a **RAG-powered PDF learning platform** using FAISS vector search and multiple LLM providers to generate document-grounded answers and AI explanations.
- Implemented PDF processing, semantic retrieval, authentication, and AI study tools including summaries, flashcards, quizzes, and learning assistance.

AI-Interviewer, Generative AI / LLM / AI Application

Technologies: Python, Flask, SQLAlchemy, Google Gemini API, Groq, OpenRouter, PyMuPDF, ReportLab, CodeMirror, Chart.js, JavaScript, HTML5, CSS3

- Developed an AI-powered mock interview platform supporting **adaptive text, voice, and coding interviews** with role-specific and resume-based questioning.
- Implemented multi-provider LLM integration, AI answer evaluation, performance scoring, readiness reports, interview integrity monitoring, authentication, and PDF report generation.

AI Resume Screening System, Generative AI / NLP / Recruitment AI

Technologies: Python, Flask, SQLAlchemy, Google Gemini API, Flask-Login, pdfplumber, ReportLab, Chart.js, Bootstrap 5, JavaScript, HTML5, CSS3

- Developed an AI recruiting platform that analyzes resumes against job descriptions using Gemini to generate **match scores, skill-gap analysis, and candidate recommendations**.
- Implemented team-based recruitment management, email verification, role-based access control, audit logging, filtering dashboards, and CSV/PDF report export.

RepoForge AI, Generative AI / Developer Tooling

Technologies: Python, Flask, SQLite, Google Gemini API, JavaScript, HTML5, CSS3, Mermaid.js

- Developed an AI-powered GitHub repository generator that creates **professional documentation, workflows, licenses, templates, and project files** using Gemini.
- Implemented hybrid AI/template generation, project history, live Markdown/Mermaid editing, ZIP export, and an AI-powered README improvement tool..

Nexus AI — AI Chatbot & Voice Assistant, LLM / Generative AI / Voice AI

Technologies: Python, Flask, Google Gemini API, Speech Recognition, Text-to-Speech, JavaScript, HTML5, CSS3, Marked.js, Highlight.js

- Developed a **Gemini-powered AI chatbot and voice assistant** supporting real-time conversations and voice-based interaction.

- Implemented multi-session chat history, Markdown and code rendering, speech recognition, text-to-speech, and voice-controlled system/web automation.

AI-Image-Generator, *Generative AI / Image Processing*

Technologies: Python, Flask, JavaScript, HTML5, CSS3, Pillow, Hugging Face API, JSON

- Developed a full-stack **AI image generation application** integrating Hugging Face models for text-to-image generation.
- Implemented prompt enhancement, image editing, history, favorites, search, sorting, and metadata management through a responsive interface.

Smart Waste Classification System, *Deep Learning / Computer Vision*

Technologies: Python, TensorFlow, Flask, OpenCV, NumPy, Keras, Chart.js

- Developed a deep learning application to classify waste into **six categories** using CNN-based image classification.
- Implemented image upload, real-time camera detection, prediction analytics, and recycling recommendations through a responsive Flask application.

 SKILLS

Programming Languages

- Python
- SQL
- JavaScript
- HTML5
- CSS3

Libraries & FrameWorks

- Flask
- TensorFlow
- Keras
- Scikit-learn
- OpenCV
- Pandas
- NumPy
- Streamlit
- PyMuPDF
- FAISS

Soft Skills

- Problem Solving
- Analytical Thinking
- Communication
- Team Collaboration
- Continuous Learning

Artificial Intelligence & Generative AI

- Large Language Models (LLMs)
- Generative AI
- Retrieval-Augmented Generation (RAG)
- Natural Language Processing (NLP)
- Prompt Engineering
- AI Chatbot Development
- Speech Processing
- Computer Vision
- Object Detection
- Image Classification

Database & Backend

- SQL
- SQLite
- PostgreSQL
- Supabase
- SQLAlchemy
- REST APIs

Machine Learning

- Machine Learning
- Deep Learning
- Supervised Learning
- Feature Engineering
- Data Preprocessing
- Model Training & Evaluation
- Predictive Modeling

AI & Development Tools

- Google Gemini API
- OpenAI API
- Anthropic API
- Groq
- OpenRouter
- Git
- GitHub
- VS Code
- Jupyter Notebook
- Google Colab
- Kaggle

 CERTIFICATES

Artificial Intelligence & Machine Learning with Python — NAVTTC

 ACHIEVEMENTS

- Developed and deployed multiple **AI, Machine Learning, Computer Vision, and full-stack Python applications**.
- Built practical AI solutions including **fraud detection, object detection, AI assistants, image generation, and intelligent automation systems**.
- Implemented modern AI technologies including **LLMs, RAG, FAISS, YOLOv8, Gemini, and Hugging Face APIs**.
- Continuously expanded expertise in **Generative AI, Machine Learning, and AI application development** through hands-on projects.